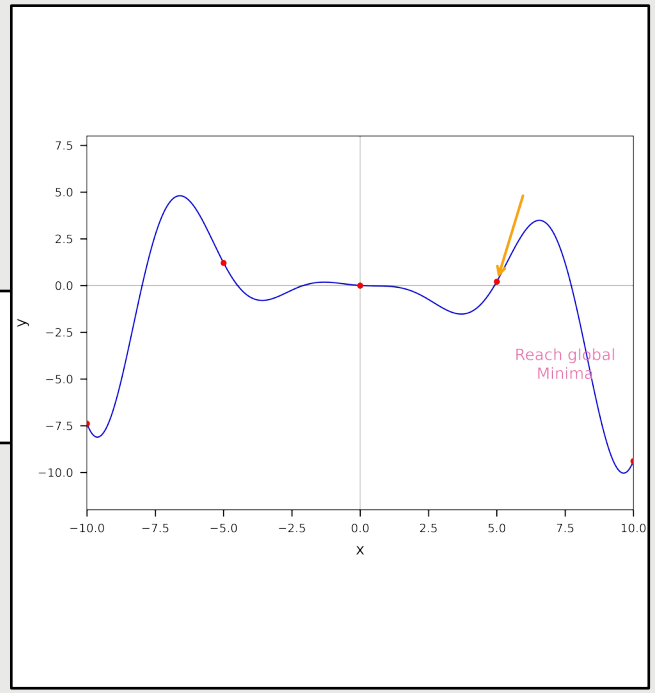
Non-Convex Function

May reach different minimum value based on **initial point and learning rate**

$$f(x) = \frac{x^2\cos(x) - x}{10}$$



step	0.2
initial X	7
delta	0.1

X is in between 9.6 and **9.8**
f(x) is in between -10.03 and -9.92

x	7	7.2	7.4	7.6	7.8	8	8.2	8.4	8.6	8.8	9	9.2	9.4	9.6	9.8	9.6	9.8	9.6
f(x)	2.99	2.43	1.66	0.69	-0.45	-1.73	-3.1	-4.5	-5.88	-7.16	-8.28	-9.17	-9.77	-10.03	-9.92	-10.03	-9.92	-10.03
x-delta	6.9	7.1	7.3	7.5	7.7	7.9	8.1	8.3	8.5	8.7	8.9	9.1	9.3	9.5	9.7	9.5	9.7	9.5
f(x-delta)	3.19	2.74	2.07	1.2	0.14	-1.08	-2.41	-3.8	-5.2	-6.54	-7.75	-8.76	-9.51	-9.95	-10.02	-9.95	-10.02	-9.95
x+delta	7.1	7.3	7.5	7.7	7.9	8.1	8.3	8.5	8.7	8.9	9.1	9.3	9.5	9.7	9.9	9.7	9.9	9.7
f(x+delta)	2.74	2.07	1.2	0.14	-1.08	-2.41	-3.8	-5.2	-6.54	-7.75	-8.76	-9.51	-9.95	-10.02	-9.7	-10.02	-9.7	-10.02
slope	-4.5	-6.7	-8.7	-10.6	-12.2	-13.3	-13.9	-14	-13.4	-12.1	-10.1	-7.5	-4.4	-0.7	3.2	-0.7	3.2	-0.7
next x	7.2	7.4	7.6	7.8	8	8.2	8.4	8.6	8.8	9	9.2	9.4	9.6	9.8	9.6	9.8	9.6	9.8